Neo Neng Kai Nigel

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Work Experience

DSO National Laboratories — Oct 2023 – Present
Member of Technical Staff (Data Scientist) — *Singapore, Singapore*

- Responsible for structuring problems to be exploited by AI, analysing current AI trends for materials and communicating and visualising results to upper management
- Led research project on inverse design for materials processing, implementing generative AI tools for microstructural generation and characterisation with verification through physics-based models

Education

Georgia Institute of Technology — Aug 2022 – Aug 2023
Master of Science in Computer Science (GPA: 4.0/4.0) — *Atlanta, Georgia, USA*

- Specialisation: Machine Learning

National University of Singapore — Aug 2018 – May 2022
Bachelor of Science (Honours) in Chemistry & Computer Science (CAP: 4.81/5.00) — *Singapore, Singapore*

- Specialisations: Materials Chemistry // Artificial Intelligence

Talks / Presentations

Lindau Nobel Laureate Conference 2022 — June 2022
Panelist - Artificial Intelligence in Chemistry — *Lindau, Bavaria, Germany*

- Discussed current ideas and future directions in artificial intelligence in chemistry with Nobel Laureates Arieh Warshel and Michael Levitt. Article available at tinyurl.com/lindau2022nigel, with video at tinyurl.com/lindau2022nigelpanel

Research Experience

Georgia Institute of Technology - CLAWS Lab — Dec 2022 – Dec 2023
Graduate Research Assistant // Affiliate — *Atlanta, Georgia, USA*

- Attached to Prof Srijan Kumar's lab for master's research project
- Curated novel datasets for the fair graph anomaly detection problem
- Benchmarked fairness regularisers, graph debiasers and graph anomaly detection models on this new dataset to understand how these methods fare and examine ways to improve them

National University of Singapore - Computational Chemistry Research Lab — Jun 2021 – May 2022
Honours Year Research Student — *Singapore*

- Attached to Prof Wong Ming Wah Richard's lab for honours year research project
- Implement and modify state-of-the-art machine learning models for chemistry (Molecule Attention Transformer) to predict sensitivity values of energetic materials from their molecular structure
- Investigated how attention matrices in the Transformer model can be used to explain predictions in a chemical context

DSO National Laboratories - Information Department — Jan 2018 – Jun 2018 // May 2020 – Aug 2020
Research Intern — *Singapore*

- Worked on the inverse problem of reconstructing chemical molecules from mass spectrometer data
- Published a paper (Chemical Structure Elucidation from Mass Spectrometry by Matching Substructures) which was accepted at NIPS 2018 Workshop: Machine Learning for Molecules and Materials
- Under Dr Chieu Hai Leong, reframed the task as a Reinforcement Learning problem, incorporating Graph Neural Networks using PyTorch Geometric and used Imitation Learning methods to jumpstart learning

Scholarships / Awards

DSTA Scholarship — *Aug 2018 – Aug 2023*

- Sponsorship of undergraduate studies and masters program

Technical Skills

- **Programming Languages**: Python, C++, C, Golang
- **Frameworks (ML / Data Science)**: PyTorch, PyTorch Geometric, Ray Tune, HuggingFace, pandas, NumPy, SciPy
- **Frameworks (Chemistry)**: RDKit, DeepChem
- **Others**: Linux, Bash, Github, Matplotlib, SikuliX